DTSC 625: Data Engineering in the Cloud

FA2 2026, Online; 3 Credits

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Important Dates

- First day of term:
  - Monday, October 19, 2026
- Live Session:
  - Tuesday, October 20, 2026 at 8:30 PM ET [Zoom Link]
- Last day to add a course:
  - Monday, October 26, 2026
- Last day to drop a course (without charge):
  - Monday, November 2, 2026
- Thanksgiving Break:
  - Monday, November 23, 2026 - Sunday, November 29, 2026
- Last day to withdraw (without academic penalty):
  - Sunday, December 13, 2026 at **5:00 PM ET**
- Last day of term:
  - Sunday, December 13, 2026
    - You can submit assignments **no later than 11:59 PM ET** on this date

Make sure to subscribe to our Student Calendar!
Not sure how to subscribe to a Google calendar? Read about it here.

Welcome to DTSC 625! This syllabus is an essential tool for navigating this course and provides comprehensive course information. It is your responsibility to familiarize yourself with its content. **There will be a quiz based on the syllabus content, and you must score 100% on this quiz to unlock certain course material.** This is to ensure that you are fully aware of the course requirements and expectations. For your convenience, a table of contents is provided on the next page to help you effectively navigate the syllabus.

Important: To access all course-related Google Docs and resources, you must log in using your Eastern email account. Some students have reported issues with accessing these resources when signed in with a personal or non-student email account. Therefore, always ensure you're logged in with your student email account.

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Course Objectives

Upon successful completion of this course, students are expected to be able to:

- Students will understand and apply fundamental data engineering principles to generate, ingest, transform, store and serve data to consumers.
- Students will be able to design and build end-to-end data pipelines using a variety of AWS services.
- Students will gain hands-on, practical experience with cloud-native data engineering tools, including AWS S3, Lambda, Step Functions, Glue, and Athena.
- Students will be able to select and justify architectural designs, articulating how a data pipeline solves a specific business problem.

Course Description

This course provides a comprehensive overview of modern data engineering principles and practices, with a specific focus on implementing these concepts within the cloud. Students will learn to design and build scalable, reliable, and cost-effective data pipelines using a variety of AWS services.

Prerequisites: DTSC-620 (Cloud Foundations)

Note: Students will use their personal AWS account and will incur charges to complete assignments.

Course Delivery

This course is delivered through Brightspace, Eastern's learning management system. You are expected to make use of the provided course materials, which include video lectures, optional non-credit quizzes, mandatory exams, and coding assignments.

As this is a self-paced course, you are responsible for managing your learning time effectively. Our expectation is that you learn using our resources, including lectures and textbook material. However, we understand that there may be times when you need additional assistance.

We recognize the value of external resources to supplement learning, and you are welcome to turn to Google, YouTube, and other outside resources for clarification or guidance. Learning how to efficiently seek help is an invaluable skill in the data science field. However, we would like to emphasize the importance of academic honesty when utilizing these external resources.

While the self-paced nature of this course affords you a degree of flexibility, it is crucial to plan and work ahead. Bear in mind that if you choose to defer work until the end of the term, no additional time will be granted for issues such as technological glitches, problems with Brightspace, email, and so forth. Extensions for *unforeseen circumstances* leading to incomplete work are discussed separately in the section Incompletes and Extensions.

Textbook Information

There is no required textbook for this course.

Recommended textbooks:

- Joe Reis and Matt Housley, *Fundamentals of Data Engineering*
- Gareth Eagar, *Data Engineering on AWS*, 2nd ed.

Additionally, you are encouraged to utilize any additional resources you find beneficial.

Eagle Learning Materials

All Eastern University students are automatically enrolled in Eagle Learning Materials, which charges graduate students a flat fee of **$125 per enrollment period** (Fall, Spring, Summer) to cover course materials for all courses taken in that period. Students have the option to opt-out, but they must do so by the course drop deadline in the first term of each enrollment period (Fall 1, Spring 1, Summer 1). **Please review the Eagle Learning Materials content in Module 0 for full program details, policies, and opt-out instructions.**
Additional information and FAQs can be found on the university's Textbooks through Eagle Learning Materials page. If you have questions, please reach out to bookstore@eastern.edu.

Technology Requirements

Course Requirements

An AWS account with administrator privileges is required. To sign up for your own account, a credit card will be needed. To ensure accurate tracking of participation and grading, students must use either their full name or their Eastern University email address as their AWS username. Accounts that do not follow this convention may not be credited for submitted work. Note: Eastern University is not liable or responsible for any charges you incur on your AWS account. It is the student's responsibility to monitor their costs and pay any costs they incur to AWS.

Honorlock Requirements

Honorlock proctoring service is used in this course, please verify that your Operating System, Browser, and Internet Speed meet Honorlock's minimum system requirements. Note that these system requirements may be different from the general program requirements outlined below. Additionally, a webcam is necessary to take exams through Honorlock.

Please see the Honorlock section of the syllabus for additional information.

General Program Requirements

Throughout the program, you will need a decent, mid-level laptop. The following specifications are recommended:

PC:

- Pentium processor or equivalent
- 4-8 GB of RAM
- 500 GB hard drive or 128 GB SSD drive
- Wireless capabilities
- Operating System: Windows 8 and above

Mac:

- At least 4 GB of RAM
- 128-256 GB hard drive
- Operating System: macOS 10.0 and above

Note: Chromebooks are not recommended for this program as they may not support some of the required software.

Please note that these specifications should be sufficient for most courses in the program. However, if you plan to work with very large datasets or require additional computational power, we may need to discuss alternative options. Please email datascience@eastern.edu or dtsc_techsupport@eastern.edu for questions about these requirements.

General Software Requirements:

Throughout the program, you will be using the following software, all of which are free:

- Jupyter Notebooks: A popular platform for interactive computing and data analysis.
- RStudio: An integrated development environment (IDE) for the R programming language.
- PostgreSQL: A powerful, open-source relational database management system.

Communication Guidelines

You are accountable for all communication sent to your @eastern.edu email account. It is important to monitor this account regularly as it will be used for class updates, registration notifications, career service announcements, and more.

Email Communications

If you need to get in touch via email, please note the following:

- *For questions about course content or assignments*, utilize the discussion boards or email the course GAs at dtsc_ga_625@eastern.edu
- *For technical inquiries or assistance with software installation, program issues, etc.,* email Tech Support at dtsc_techsupport@eastern.edu.
- *For general program inquiries*, email datascience@eastern.edu.
- *For academic advising or to request an extension*, contact dsadvising@eastern.edu.
- *For personal inquiries or concerns*, reach out directly to the course instructor.

GA Communication

The primary means of GA communication will be through discussion boards, where you can interact with both Graduate Assistants (GAs) and your peers. GAs will make every effort to respond to your queries within 1-2 business days, and other students are also welcome to answer questions.

We encourage you to initially use the appropriate discussion board for most course content questions. However, there may be instances where email communication is necessary. **It is important to note that the discussion board should not be used to post code, assignment solutions, personal queries, or questions about grading.** These matters should be addressed directly via the GA email. Questions emailed to the GA mailbox are usually responded to within 1-2 business days.

When contacting via email, please format your subject line with your Last Name and the nature of your email:

"LastName - Question on Assignment #"

*This includes emails sent through Brightspace, where you will need to modify the default subject line.

Why do we require your last name? We utilize Google groups for communication, and due to its grouping mechanism, emails with the same subject line—regardless of the sender—are grouped together. For this reason, and to ensure we promptly address your questions, we need you to include your last name in the subject line.

GA Scope

The GAs are available to assist with course content, assignments, quizzes, and more. It is important to be aware of the scope of the support they offer. They are here to clarify specific questions related to course material and will not provide answers. To help you best, it's important that your questions are clear and specific, rather than general or vague. Additionally, they won't pre-grade assignments or address questions about an assignment's correctness. Their primary role is to guide and assist you in understanding the concepts so that you can apply them independently.

Our GAs are here to assist all students in the course and we ask that you be mindful of the amount of time you are requesting from them. If you find yourself frequently requesting assistance from the GAs, we suggest connecting with the course instructor to help you identify strategies to troubleshoot independently and enhance skills like debugging and problem-solving.

Course Details

The Data Science program courses all follow a general common framework, but each course has its unique requirements and expectations. Below you'll find any course-specific items you should be aware of. By familiarizing yourself with these details, you ensure that you are prepared for the unique aspects of this course. You must read through the following sections thoroughly as this material will be covered in the mandatory Syllabus Quiz. Please note that you must achieve a score of 100% on this quiz to unlock further course content.

Course Outline

The following course outline provides an overview of the structure and content of this course. It lists the general content in this course. This outline functions as a general summary to help you plan your studies effectively.

- Course and Instructor Introduction
    - Course Overview
    - Course Introduction
    - Instructor Introduction
    - Course Expectations
    - Syllabus
- Intro to Data Engineering
- Governance - Security, Privacy & Cataloging
- Data Generation and Ingestion
- Data Storage
- Queries, Modeling, Transformation
- Orchestration
- Serving Data to Consumers
- (Optional) Snowflake

Grading

Exams (30%)

You will have a midterm (10%) and final exam (20%) in the course, assigned on Brightspace, worth a total of 30% of your grade. The exams are timed at **120 minutes**. If you have documented accommodations for more time, please reach out to the professor. **The use of any resources such as notes, textbooks, LLMs, internet sites, etc., during exams is not**

permitted. The format for these exams will include a variety of question types such as multiple choice, multiple select, fill-in-the-blank, and short-answer questions.

You will be able to take each exam **three** times**.** Exams will consist of a random pool of questions; therefore, each time you take the exam you should expect to see different questions.

Before you can attempt the first exam, **you must earn a 100% on the Course Overview and Syllabus quiz**. This quiz does not contribute towards your final grade and you have unlimited attempts to achieve the required 100% to move on.

After completing your exam, you will be able to review the questions you answered correctly and incorrectly. However, when you choose to view your submission, you will only have 15 minutes to do so. This helps prevent exam questions from becoming public, while also helping you understand where your answers were correct or incorrect. Please ensure you view your submission when undistracted.

HonorLock

Honorlock will proctor your exams this semester. Honorlock is an online proctoring service that allows you to take your exam from the comfort of your home. You DO NOT need to create an account or schedule an appointment in advance. Honorlock is available 24/7, and all that is required is a computer, a working webcam, a microphone, and a stable internet connection.

To get started, you will need Google Chrome and download the Honorlock Chrome Extension.

When you are ready to complete your assessment, log into your LMS, go to your course, and click on your exam. Clicking "Launch Proctoring" will begin the Honorlock authentication process. Honorlock will be recording your exam session through your webcam and microphone and recording your screen. Honorlock also has an integrity algorithm that can detect search-engine use, so please do not attempt to search for answers, even if it's on a secondary device. **Honorlock proctoring violations may result in grade penalties.**

The following are **prohibited behaviors** and may result in failing the assignment or the course:

- Typing on a different computer, device, or browser while taking an exam
- Accessing course notes while taking an exam
- Accessing websites or LLMs (like ChatGPT, Gemini, etc.) while taking an exam

Honorlock support is available 24/7/365. If you encounter any issues, you may contact them through live chat on the support page or within the exam itself. Some guides you should review are Honorlock MSRs, Honorlock FAQs for Test Takers, Honorlock Knowledge Base, and How to Use Honorlock. Good luck!

We understand that you may have concerns when taking exams remotely and with that comes our assurance of privacy protection. You can read Honorlock's student privacy statement **here**.

Honorlock Guidelines

- Photo ID Verification **Not Required**
- Workspace Scan Photo **Not Required**
- Open Book Not Allowed
- Notes Not Allowed
- Scratch Paper Not Allowed
- Restroom Breaks Not Allowed
- Handheld Calculator Not Allowed
- Headphones Not Allowed
- Hats Allowed
- Public Area Not Allowed
- Background Noise Allowed
- Mobile Phone/Smart Watch Use Not Allowed

Assignments (70%)

You will follow lab directions and take screenshots of various parts of your AWS management console and paste these into a template. Assignments are hand-graded and are one submission only. Be sure to include all the requested information indicated in the template (such as your name in your AWS environment). Use either your full name or your Eastern University email address as your AWS username. Work submitted under any other username cannot be matched to your record and may not receive credit.

Since students complete assignments in their own AWS account, troubleshooting issues with GAs or the professor may be challenging. Because of this, we encourage the use of LLMs to assist in your troubleshooting.

Grade Distribution

Graded Task	Grade Weight
Midterm	10%
Final Exam	20%
Exams Total	30%
Create AWS Budgets	5%
Lambda CSV to Parquet	5%
Lake Formation	10%
Ingestion with Kinesis	10%
Joining and Cleaning Data with Glue Studio	10%
Storage Optimization and Data Cleaning with Step Functions	10%
Step Functions with Athena Queries	15%
Cost Estimate for St. Gregory's	5%
Assignments Total	70%
Total	100%

Grade Scale

93+	A
90	A-
87	B+
83	B
80	B-
77	C+
73	C
70	C-
<70	F

Student Responsibilities

As a student in this course, you are expected to adhere to certain responsibilities that are outlined below. It is imperative that you carefully read and fully understand these items as they ensure your academic success and maintain the integrity of our program.

- You are required to retain copies of all materials submitted for grading purposes. In the unlikely event of a misplaced assessment, you will be responsible for supplying a copy. This ensures the preservation of your academic efforts and record.

- You are expected to adhere to the standards of academic honesty in all aspects of this course. Collaboration that leads to sharing or copying of code is strictly prohibited, as is copying answers found online. All code submitted for grading must be your own original work. The code you submit must reflect your own understanding and effort.

- You are required to maintain the confidentiality of the course materials. Posting solutions or assignment materials publicly, whether on online forums, personal blogs, or code repositories such as GitHub, is not allowed. If you use GitHub or similar platforms to post course related content, ensure that your repository is set to private.

Your commitment to these responsibilities help you set a benchmark for the level of professional conduct expected in the Data Science industry.

For additional information, see the Academic Dishonesty Policy outlined here.

Academic Policies

Please see Eastern University's **University and Academic Policies** for the most recent college-wide policies. Students can also find detailed statements printed in the College Catalog. For additional information or context-specific definitions of these policies, please utilize the Student and Instructor Handbooks of the respective program or department offering this course.

Academic Dishonesty

It is the student's responsibility to familiarize themselves with acceptable standards for research and documentation and adhere to them. The definition of academic dishonesty and the corresponding penalties are clearly outlined in the Catalog.

Students are strictly prohibited from posting or sharing solutions or answers to any assignments, quizzes, exams, projects, or any other items that contribute to the course grade. This includes posting solutions on public GitHub repositories or similar platforms. If students wish to build a portfolio, they should create a private repository for sharing purposes. Violation of this policy can result in penalties ranging from failing a class to dismissal from the program.

We expect all work conducted at Eastern University to reflect honest and ethical behavior. The primary rule for ensuring academic integrity is that all submitted work must be original and produced by the individual student.

To uphold academic integrity, it is not permitted to view another student's work prior to submitting your own, nor should you provide your work to someone who has not yet submitted their work. Collaborating with other students is only allowed when seeking clarification on a topic or assignment instructions. Collaborating on quizzes and exams is not permitted.

Do's and Don'ts include, but are not limited to:

Do:

- Give credit when using other’s work in both written and coding assignments.
- Use a few lines of code from online sources *only if proper credit is given and it does not provide a complete solution to an assignment*.
- Assist other students in understanding assignment instructions or specific topics.
- Reach out to other students on the discussion board to ask clarifying questions (without showing your code).
- Reach out to your course Graduate Assistant or Instructor for clarification or assistance when needed.
- Ensure that you give yourself enough time to work on assignments to reduce the temptation to look up answers.
- Utilize outside resources solely for the purpose of seeking clarification on topics, obtaining assistance with specific error messages, understanding the functionality of code, and similar purposes that enhance your comprehension.

Don't:

- Post assignment or exam solutions online where they can be accessed and viewed by others.
- Submit someone else's writing or code as your own.

- Share or make your code available to other students who have not yet submitted their assignments.
- Collaboratively work on an assignment by dividing tasks and combining them before submission. All work must be individually completed.
- Review another student's code or search for a complete solution online before submitting your assignment.
- Fabricate results, data, or statistics.
- Share or discuss specific exam questions publicly or with other students (except when discussing with the course instructor and GAs, which is appropriate).
- Submit work that has been previously used in another course or submitted to another university program (especially for your capstone project).

It is the student's responsibility to understand acceptable research and documentation standards and abide by them. If you are unsure whether something is acceptable, please discuss with the appropriate staff member beforehand. All codegrade submissions are subject to audit, and credit may be deducted if assignments do not adhere to the provided instructions and the guidelines outlined here.

The bottom line is that ALL work must be your own.

AI and Large Language Model (LLM) Use

In our MS in Data Science and Data Analytics programs, students are encouraged to utilize AI and Large Language Models (LLMs) ethically to support their learning. Students may use LLMs for general coding assistance, understanding theoretical concepts, debugging, and studying. **However, using LLMs to complete assignments, assessments, or to generate specific solutions for graded work is strictly prohibited.** Plagiarism through LLMs is also not allowed. Violations of this policy can result in severe disciplinary actions, including a failing grade or dismissal from the program. For any uncertainties about acceptable use, students should consult their instructor or program staff.

Do:

- Make use of AI-based tools from third-party providers (like ChatGPT, GitHub Copilot, the new Bing, etc.) to assist in studying for an exam or helping to understand a concept more deeply.

Don't:

- Make use of AI-based tools from third-party providers (like ChatGPT, GitHub Copilot, the new Bing, etc.) to assist in generating answers or writing code

Please see the Acceptable Use of AI and LLMs Student Policy here for further information.

Class Attendance Policy

This course is asynchronous and online, meaning there is no formal time to meet. Therefore, attendance is not mandatory. However, periodic activity checks may be conducted to monitor participation. If you receive an email regarding this, please understand that it is for your benefit.

Incompletes and Extensions

Each request for an Incomplete grade (or course extension) is taken seriously and reviewed meticulously. Requests are granted strictly under "extreme and unforeseen extenuating circumstances" for students who are unable to complete the course on time. **To be eligible for an Incomplete, students must have already completed at least 75% of graded course elements (exams, assignments, quizzes, etc).**

Note: Throughout the program, students are limited to a maximum of two Incompletes, and they cannot be requested in consecutive terms.

The procedure for seeking a course extension requires students to first contact their **academic advisor** (dsadvising@eastern.edu) detailing the reason for their request. Please keep in mind that, unless there are extenuating circumstances, these requests must be received by the academic advisor and approved by your course instructor *before* the term ends.

Please consult the Student Handbook found on the "Master's in Data Science Program" Brightspace page for the complete overview of this policy.

Student Disability Policy

Notice to Students with Physical or Learning Disabilities: To be eligible for disability accommodations at Eastern University, students must submit a written request to the Cushing Center for Counseling and Academic Support (CCAS) and be deemed eligible for such accommodations. For this course, the professor must receive a written request from CCAS to implement any accommodations. Please note that disability accommodations are not retroactive and will not be applied until a request from CCAS is received. All accommodations need to be reactivated by the student prior to the start of each new academic session.

The CCAS is now 3 distinct offices:

- CAPS (Counseling and Psychological Services)
    - Contact: caps@eastern.edu or 610-341-5837
    - Director: Lisa Hemlick
    - Services
        - Counseling and mental health support
        - College Success program for students on the Autism Spectrum (CSP)

- ARC (Academic Resource Center)
    - Contact: arc@eastern.edu or 610-225-5567
    - Director: John Nordlof
    - Services:
        - Tutoring
        - Writing Support
        - Academic Coaching
- ODA (Office of Disability and Access)
    - Contact: oda@eastern.edu
    - Director: Lori Dziedziak
    - Services:
        - Students who need accommodations for diagnosed disabilities and medical conditions

For further information, please visit: https://www.eastern.edu/student-support